



25801831

QP CODE: 25801831

Reg No :

Name :

M.C.A DEGREE EXAMINATION, APRIL 2025

Second Semester

MASTER OF COMPUTER APPLICATION

CORE - MCACT201 - OPTIMIZATION TECHNIQUES FOR COMPUTER APPLICATIONS

2020 Admission Onwards

DA2ACD08

Time: 3 Hours

Maximum: 75 Marks

Part A

*Answer any **ten** questions*

*Each question carries **3** marks*

1. Write the given LPP in standard form : $\text{Min } Z = x + 4y$ subject to $x - 3y \geq 10$, $2x + y = 6$, $x, y \geq 0$
2. Explain mathematical modelling of a Linear Programming Problem.
3. How do you interpret Slack and Surplus variable ?
4. What is the criterion for selecting the entering basic variable?
5. Define Transportation Problem.
6. Give the mathematical formulation of an assignment problem.
7. Formulate the travelling salesman problem as an Assignment problem.
8. What is meant by pay off matrix?
9. What is meant by principle of dominance?
10. Explain the following terms (i) Earliest event time (ii) Latest event time.
11. What is meant by simulation ? Why is it used ?





12. What is meant by 'no passing rule' in sequencing problem ?

(10×3=30 marks)

Part B

Answer **all** questions

Each question carries **9** marks

13. a) A company produces two types of hats. Each hat of the first type requires twice as much labour time as the second type. If all hats are of the second type only, the company can produce a total of 500 hats a day. The market limits daily sales of the first and second types to 150 and 250 hats. Assuming the profit per hat is Rs. 8 for type 1 and Rs. 5 for type 2, formulate the problem as a linear programming model in order to determine the number of hats to be produced of each type so as to maximise the profit.

OR

- b) Solve the following LPP using graphical method:- Max $P = 2x + 3y$
Subject to $x + y \leq 30$, $y \geq 5$, $0 \leq y \leq 15$,
 $0 \leq x \leq 20$, $x - y \geq 0$

14. a) Apply the simplex algorithm to solve the following linear model
Minimize $Z = x_1 - 3x_2 + 2x_3$
subject to $3x_1 - x_2 + 2x_3 \leq 9$, $-2x_1 + 4x_2 + x_3 \leq 14$, $-4x_1 + 4x_2 + 8x_3 \leq 10$, $x_1, x_2, x_3 \geq 0$

OR

- b) Maximize $Z = x_1 + 5x_2$ subject to
 $3x_1 + 4x_2 \leq 6$, $x_1 + 3x_2 \geq 2$, $x_1, x_2 \geq 0$
Use Big M method to solve the above LP Model.

15. a) Solve the Transportation Problem

Warehouse	Market				Supply
	A	B	C	D	
1	6	3	5	4	22
2	5	9	2	7	15
3	5	7	8	6	8
Demand	7	12	17	9	

OR





- b) Find an assignment programme which will maximise the total profit

Instructors	Courses			
	A	B	C	D
1	6	4	6	7
2	2	6	5	7
3	2	3	4	6
4	3	1	3	4

16. a) Solve

Player A	Player B				
	2	-4	6	-3	5
	-3	4	-4	1	0

OR

- b) The belt snapping for conveyers in open cast mine occur at a rate of 2 per shift. There is only one hot plate available for vulcanising and it can vulcanise on an average 5 belts snap per shift. (i) What is the probability that when a belt snaps, the hot plate is readily available? (ii) What is the average number in the system? (iii) What is the waiting time of an arrival? (iv) What is the average waiting time plus vulcanising time?
17. a) A small project is composed of seven activities. The time estimates are listed in the table as follows:

	Activity	1-2	1-3	2-4	2-5	3-5	4-6	5-6
Time estimates(weeks)	Optimistic(t_0)	1	1	2	1	2	2	3
	Most Likely(t_m)	1	4	2	1	5	5	6
	Pessimistic(t_p)	7	7	8	1	14	8	15

- 1) Draw the project network
- 2) Find the expected duration and variance of each activity
- 3) Find the critical path and project duration
- 4) What is the probability that the project will be completed 4 weeks earlier than expected

OR





- b) A tourist car operator finds that during the past few months, the car's use has varied so much that the cost of maintaining the car varied considerably. During the past 200 weeks the demand for the car fluctuated as shown below:

Trips/week	0	1	2	3	4	5
Number of days(frequency)	16	24	30	60	40	30

Using random numbers simulate the demand for a 10 week period

[Use the random numbers:85,96,18,96,10,84,56,11,52,02]

(5×9=45 marks)

